

Week 1: Literature Review

Analysing the Role of Big Data Analytics for Cyber Threat
Detection and Prevention in Banking Organisations of Ireland

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15 Peer-Reviewed Papers (2025-2026)

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1. Introduction

This literature review presents 15 recent peer-reviewed papers published between 2025 and 2026 that are directly relevant to the research topic: 'Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland'. The review covers five key thematic areas:

- (a) Big Data Analytics in Cybersecurity
- (b) Machine Learning for Cyber Threat Detection
- (c) Intrusion Detection Systems (IDS)
- (d) Fraud Detection in Banking using ML
- (e) Predictive Analytics and Cyber Threat Intelligence in Financial Services

Each paper is summarised with its key findings, methodology, and relevance to the current study. This review builds upon the earlier literature review (CA1 proposal) and adds the most recent academic contributions to establish a comprehensive foundation for the research.

2. Big Data Analytics in Cybersecurity

Paper 1

BankNet: Real-Time Big Data Analytics for Secure Internet Banking

Authors: Sathupadi, K., Achar, S., Bhaskaran, S.V. et al.

Year: 2025 | Journal: Big Data and Cognitive Computing (MDPI), Vol. 9, No. 2, p. 24

Summary: This study proposes BankNet, a real-time big data analytics framework designed for secure internet banking. The framework integrates big data processing pipelines with machine learning-based anomaly detection to identify suspicious transactions in real time. The paper demonstrates that combining distributed data processing with intelligent threat detection significantly reduces response times to cyber threats in banking environments. This paper is highly relevant to the current study as it directly addresses the application of big data analytics in banking cybersecurity.

Paper 2

Big Data in Cybersecurity: Enhancing Threat Detection with AI and ML

Authors: Kumar, B.H., Nuka, S.T., Malempati, M. et al.

Year: 2025 | Journal: Metallurgical and Materials Engineering

Summary: This paper examines how big data technologies can be leveraged alongside AI and machine learning to enhance cyber threat detection. The study reviews various big data frameworks such as Hadoop and Spark and evaluates their effectiveness in processing large-scale security data. The authors conclude that big data-driven approaches significantly outperform traditional rule-based systems in detecting advanced persistent threats and zero-day attacks. The findings support the use of big data analytics as a core component of modern cybersecurity infrastructure.

Paper 3

Enhancing Cybersecurity Risk Management in Fintech Through Advanced Analytics and Machine Learning

Authors: Kokogho, E., Okon, R., Omowole, B.M. et al.

Year: 2025 | Journal: Frontiers in Science

Summary: This study investigates the role of advanced analytics and machine learning in managing cybersecurity risks within fintech organisations. The research highlights that fintech companies face unique security challenges due to the

volume of digital transactions they process. The paper proposes a risk management framework that uses predictive analytics to identify and mitigate cyber threats before they cause financial damage. The study is relevant to Irish banking organisations that are increasingly adopting fintech solutions.

3. Machine Learning for Cyber Threat Detection

Paper 4

Machine Learning-Driven Approaches to Cyber Threat Detection: Enhancing IoT Security

Authors: Ali, M., Raza, A., Akram, M.A., Arif, H., Ali, A.

Year: 2025 | Journal: *Journal of Informatics and IT (acsit.org)*

Summary: This paper presents machine learning-driven approaches for cyber threat detection with a focus on IoT security. The study evaluates several ML algorithms including Random Forest, SVM, and gradient boosting for their effectiveness in detecting cyber threats across IoT networks. The results show that ensemble methods provide the highest detection accuracy. The paper contributes to understanding how ML techniques can be adapted for detecting threats in interconnected banking systems that rely on IoT devices.

Paper 5

Machine Learning-Based Anomaly Detection for Cyber Threat Prevention

Authors: Mizanur, M., Kumer, S., Reza, N.

Year: 2025 | Journal: *Journal of Primeasia (emanresearch.org)*

Summary: This study explores machine learning-based anomaly detection techniques for preventing cyber threats. The paper compares supervised and unsupervised learning approaches for identifying anomalous network behaviour. The results demonstrate that hybrid approaches combining both supervised classification and unsupervised anomaly detection yield the best performance in terms of precision and recall. This work has been cited 37 times, indicating its significance in the field.

Paper 6

Adversarial Machine Learning Framework for Robust Banking Security: The Automated Cyber Threat Detection and Prevention (ACTP) Tool

Authors: Padmanaban, H., Sharma, Y.K., Sharma, P. et al.

Year: 2026 | Journal: *Journal of The Institution of Engineers (India)*, Springer

Summary: This paper introduces the ACTP tool, an adversarial machine learning framework designed specifically for robust banking security. The framework addresses the challenge of adversarial attacks on ML-based security systems in banking. The authors demonstrate that adversarial training improves model robustness against evasion attacks while maintaining high detection accuracy. This is one of the most directly relevant papers to the current study as it focuses on banking-specific cyber threat detection.

4. Intrusion Detection Systems (IDS)

Paper 7

A Comprehensive Survey on Intrusion Detection Systems with Advances in Machine Learning, Deep Learning and Emerging Cybersecurity Challenges

Authors: Hozouri, A., Mirzaei, A., Effatparvar, M.

Year: 2025 | Journal: *Discover Artificial Intelligence*, Springer

Summary: This highly cited survey (181 citations) provides a comprehensive review of intrusion detection systems incorporating machine learning and deep learning techniques. The paper analyses the evolution of IDS from signature-based to anomaly-based detection and discusses emerging challenges including encrypted traffic analysis, adversarial attacks, and concept drift. The survey identifies Random Forest and deep neural networks as the most effective algorithms for network intrusion detection, which is directly relevant to the ML models being evaluated in this study.

Paper 8

An Advanced Cyber Security Model Using Federated Machine Learning Approach for Intrusion Detection in Networks

Authors: Laddi, M., Allagi, S., Rachh, R. et al.

Year: 2025 | Journal: Journal of Cloud Computing and Engineering (bonviewpress.com)

Summary: This paper proposes an advanced cybersecurity model that uses federated machine learning for intrusion detection. The federated approach allows multiple organisations to collaboratively train IDS models without sharing sensitive data. The study demonstrates improved detection rates for distributed denial of service (DDoS) attacks and data exfiltration attempts. This approach is particularly relevant for banking organisations in Ireland that need to share threat intelligence while complying with GDPR.

Paper 9

Development of an Intelligent Intrusion Detection System for IoT Networks Using Deep Learning

Authors: Zhang, H.

Year: 2025 | Journal: Discover Internet of Things, Springer

Summary: This study develops an intelligent IDS using deep learning techniques for IoT network environments. The paper employs convolutional neural networks (CNN) and recurrent neural networks (RNN) architectures to detect network intrusions in real time. The model achieves high accuracy on benchmark datasets including CICIDS and NSL-KDD. With 43 citations, this paper validates the use of deep learning for IDS development, supporting the broader investigation of ML-based security in banking networks.

5. Fraud Detection in Banking Using Machine Learning

Paper 10

Machine Learning-Based Fraud Detection Systems and Their Effectiveness in Reducing Cybersecurity Risks in Digital Banking

Authors: Naeem, W., Butt, M.A., Javeid, U.

Year: 2026 | Journal: Social Science Review Archives

Summary: This paper evaluates the effectiveness of machine learning-based fraud detection systems in reducing cybersecurity risks in digital banking. The study compares Decision Tree, Random Forest, and SVM models for detecting fraudulent transactions. The results show that Random Forest achieves the highest accuracy and F1-score among the three models. This paper is directly aligned with the current study as it uses the same three ML algorithms being investigated.

Paper 11

Early Detection of Digital Transaction Fraud in Banking Systems Using the Random Forest Algorithm

Authors: Dhammayogi, M.B.D., Pramatha, C.R.A.

Year: 2026 | Journal: Journal of Information Technology and Computer Science (JITECS)

Summary: This study focuses on the early detection of digital transaction fraud in banking using the Random Forest algorithm. The paper demonstrates that Random Forest can effectively identify fraudulent patterns in banking transaction data with high precision and recall. The research uses real-world banking transaction data and highlights the importance of feature engineering in improving model performance. This paper provides useful benchmarks for the Random Forest model evaluation in the current study.

Paper 12

Enhanced Banking Fraud Detection: A Comparative Analysis of Supervised Machine Learning Algorithms

Authors: Sanda, A.M., Mukhtar, U.S. et al.

Year: 2026 | Journal: Proceedings of Engineering and Technology (FETICON)

Summary: This paper presents a comparative analysis of supervised machine learning algorithms for banking fraud detection. The study evaluates multiple algorithms including Decision Tree, Random Forest, SVM, and logistic regression on banking fraud datasets. The results confirm that ensemble methods, particularly Random Forest, provide the best balance between detection accuracy and computational efficiency. The comparative methodology used in this paper is similar to the approach adopted in the current study.

6. Predictive Analytics and Cyber Threat Intelligence

Paper 13

The Role of Threat Intelligence in Preventing Financially Motivated Cyberattacks

Authors: Eshra, S.A., Zohora, F.T., Akter, S., Rasul, I. et al.

Year: 2025 | Journal: Cyber Intelligence Review (jecir.com)

Summary: This study examines the role of cyber threat intelligence (CTI) in preventing financially motivated cyberattacks targeting banking and financial institutions. The paper argues that integrating threat intelligence feeds with machine learning models enables proactive threat detection and response. The study highlights that CTI-driven security frameworks can reduce the average time to detect and respond to financial cyberattacks. This is relevant to Irish banking organisations seeking to adopt proactive cybersecurity strategies.

Paper 14

Risk Assessment Framework for US Banking Institutions: Integrating Predictive Financial Analytics, Regulatory-Aware Governance and Cybersecurity Risk Intelligence

Authors: Ullah, N., Ishaq, M., Rahman, F., Aslam, M.U. et al.

Year: 2026 | Journal: Journal of Management Science Research Reports

Summary: This paper proposes a comprehensive risk assessment framework for banking institutions that integrates predictive financial analytics with cybersecurity risk intelligence. The framework combines big data analytics, predictive modelling, and regulatory compliance to provide a holistic approach to banking security. The study demonstrates that predictive analytics can identify emerging threats with higher accuracy than traditional risk assessment methods. Although focused on US banking, the framework is applicable to Irish banking organisations facing similar regulatory and security challenges.

Paper 15

Artificial Intelligence and Cybersecurity in Banking Sector: Opportunities and Risks

Authors: Kovacevic, A., Radenkovic, S. et al.

Year: 2025 | Journal: *Contemporary Issues in...* (University of Belgrade Repository)

Summary: This study analyses the opportunities and risks associated with using artificial intelligence for cybersecurity in the banking sector. The paper discusses how AI-powered security tools can enhance threat detection, fraud prevention, and regulatory compliance in banking organisations. However, it also warns about risks including adversarial attacks on AI models, data privacy concerns, and the need for skilled personnel. The paper provides a balanced view of AI adoption in banking cybersecurity, which aligns with the current study's objective of identifying challenges in implementing big data analytics in Irish banking organisations.

7. Summary of Reviewed Papers

#	Authors	Year	Journal	Key Focus
1	Sathupadi et al.	2025	Big Data & Cog. Computing	Real-time banking security
2	Kumar et al.	2025	Metallurgical & Mat. Eng.	Big data + AI threat detection
3	Kokogho et al.	2025	Frontiers in Science	Fintech risk management
4	Ali et al.	2025	J. Informatics and IT	ML for IoT cyber threats
5	Mizanur et al.	2025	J. Primeasia	Anomaly detection for threats
6	Padmanaban et al.	2026	J. Inst. Engineers (Springer)	Adversarial ML banking tool
7	Hozouri et al.	2025	Discover AI (Springer)	IDS survey: ML & DL
8	Laddi et al.	2025	J. Cloud Computing & Eng.	Federated ML for IDS
9	Zhang, H.	2025	Discover IoT (Springer)	Deep learning IDS
10	Naeem et al.	2026	Social Sci. Review Archives	ML fraud detection banking
11	Dhammayogi et al.	2026	JITECS	Random Forest fraud detect.
12	Sanda et al.	2026	FETICON Proceedings	Comparative ML for fraud
13	Eshra et al.	2025	Cyber Intelligence Review	Threat intelligence finance
14	Ullah et al.	2026	J. Mgmt Sci. Research Rep.	Predictive analytics banking
15	Kovacevic et al.	2025	Univ. of Belgrade	AI & cybersecurity banking

8. Key Findings from the Literature

1. Big Data Analytics significantly enhances cyber threat detection in banking by enabling real-time processing of large volumes of transaction and security data (Papers 1, 2, 3).
2. Machine learning algorithms, particularly ensemble methods like Random Forest, consistently outperform traditional rule-based security systems in detecting fraudulent and suspicious activities (Papers 4, 5, 6, 10, 11, 12).
3. Among the three ML models relevant to this study (Decision Tree, Random Forest, SVM), Random Forest achieves the best overall performance in terms of accuracy, precision, recall and F1-score for fraud detection in banking (Papers 10, 11, 12).
4. Modern intrusion detection systems benefit from deep learning and federated learning approaches that improve detection while preserving data privacy (Papers 7, 8, 9).
5. Predictive analytics and cyber threat intelligence enable proactive threat detection, allowing financial institutions to identify and respond to threats before damage occurs (Papers 13, 14, 15).
6. Key challenges include data privacy and GDPR compliance, adversarial attacks on ML models, class imbalance in fraud datasets, and the need for skilled cybersecurity personnel (Papers 3, 8, 15).

9. References

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